

Sea Water Material Manual

email: hermanmichael94@gmail.com

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1. Brief

It is in my best interest that this shader will help you in your game development journey. This material focuses on appearance rather than precision and realism. So pardon me for any imperfection or inconveniences this product brought you, and of course it will be improved going forward. Beforehand, If this manual gives unclear explanation on something, don't hesitate to email me.

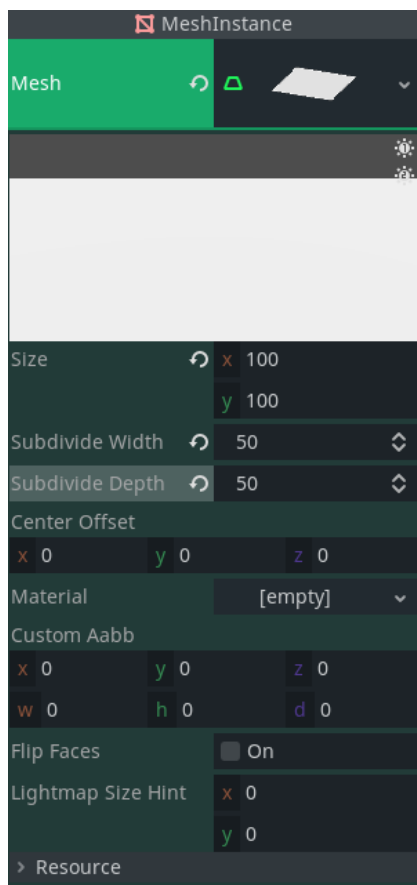
2. Folder Structure

Let's start by familiarizing with our folder structures

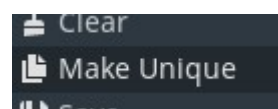
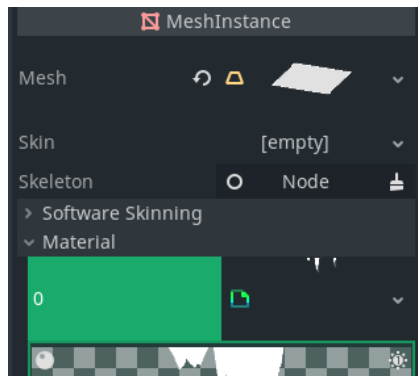
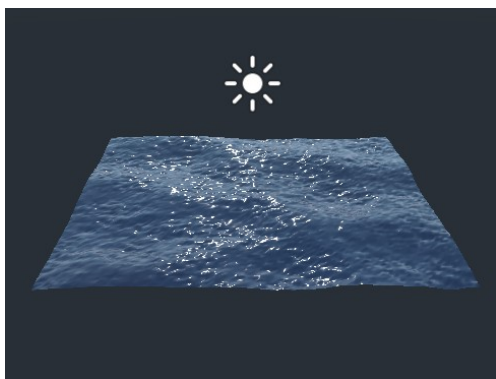
Res://			
	Demo		Contains demo files of all variance possible use case
	Materials		
		Script	Contains toolscript to aid shader setup
		Sea_Water_A	Contains Variant Material
		Sea_Water_B	
		Sea_Water_C	
		Sea_Water_D	Texture folder inside are important part of the shader, make sure to include them in your project
	Miscellaneous		
		Mesh	Mesh asset used for demo
		Textures	Contains sky texture for demo

3. How to Use

This material only works well with plane mesh, so first let's make **MeshInstance** Node and click on **Mesh** properties → **New PlaneMesh**. As for the size and subdivision of the plane you are free to choose, though I advice for a large seawater scenario to use a size to subdivision ratio by 1 to 1 or 2 to 1 for an acceptable look. The more the subdivision the better, but remind you again of the performance.



Next, on the Mesh Instance's own **Material** properties → **Load**, then navigate to the **Materials** → **Sea_Water_A** Folder, and finally choose the "**Sea_Water_A.material**" or simply drag-and-drop the corresponding material resource from file manager to the inspector. Don't forget to **Make Unique** so that it won't duplicate any changes you made to all owners of this material.



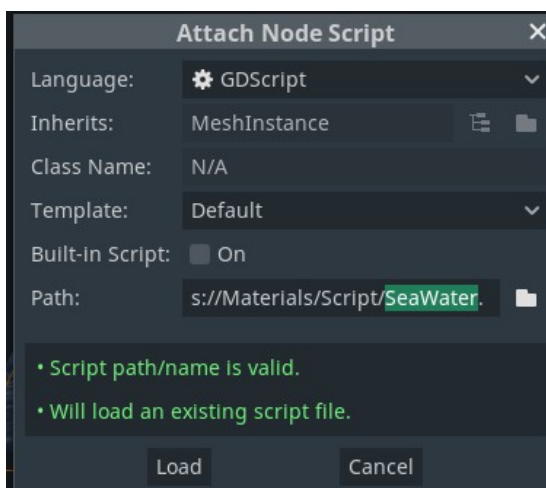
At this point, you can use it as it is or play with the **shader param**. Steps beyond here are not mandatory.

4. Toolscript Helper

Godot generously provide us with helpful feature that allows us to run gdscrip inside the editor without the need to run the program. This can be used for passing parameters from script and observe the changes in editor. For this case, we use toolscript for two functions:

A. Point Of Influence

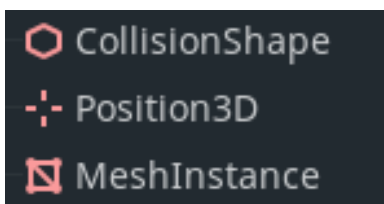
This will add influencing forces so that it can bend wave direction and height. To use, attach the **Sea_Water.gd** to the **MeshInstance**.



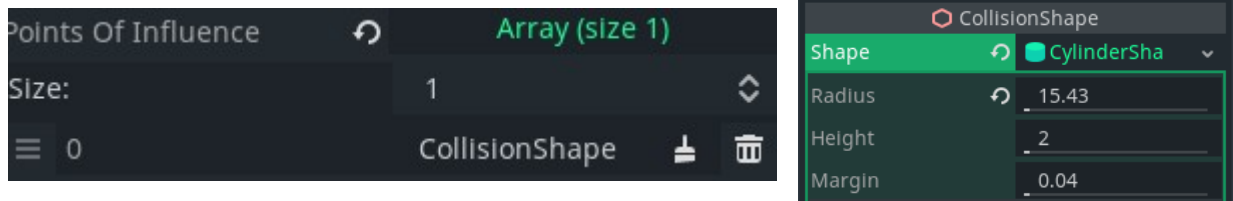
On the inspector panel, under **script variable** we can see some exported options. The one that we interested in is the point of influence array. Add few slots there for the spatial data.



This array only accepts child of **Spatial** type node. So as long as the node has a transformation property, the script will blindly consume it. That being said, I recommend to use **CollisionShape** type or some of these types.



The reason is that this script use **radius** attribute as influence factor; but if no attribute exist then it will use other type of approximation such as **AABB** dimension or simply just a hard coded constant. But for now we will use **CollisionShape** type as the influence.

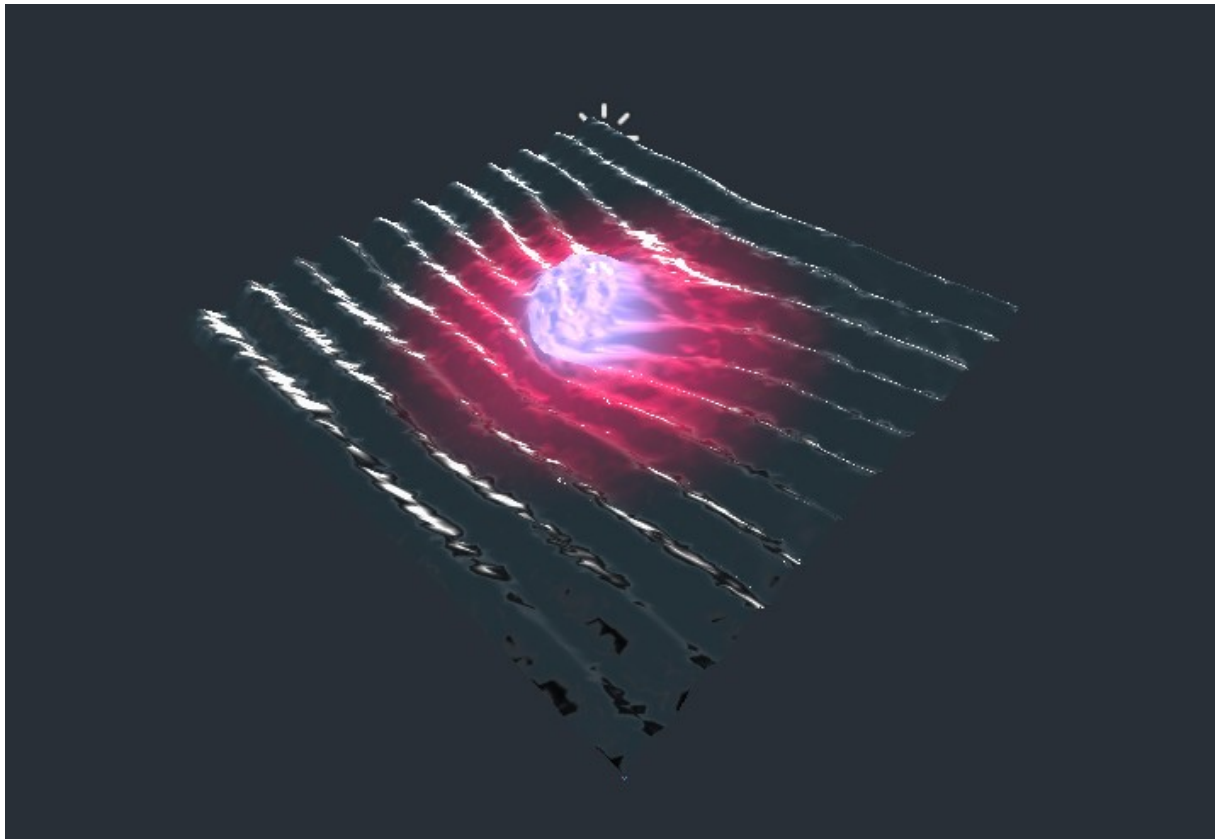


Save your scene, close, then reopen again. If successful, output console should have given similar message like this:

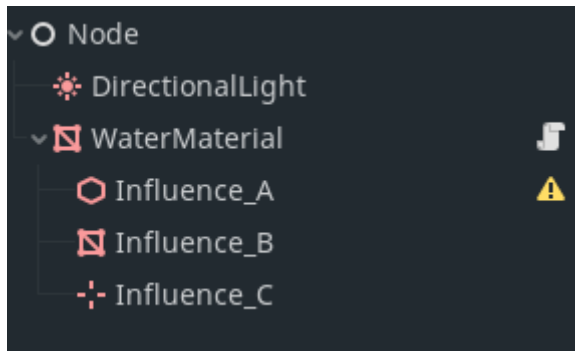
```
Object : CollisionShape, Add Locations : (0, 0, 0)  
Object : CollisionShape, set radius : 15.43
```

This step is necessary because our script doesn't use real-time update. So make sure you do close and restart whenever you update influence data or other options.

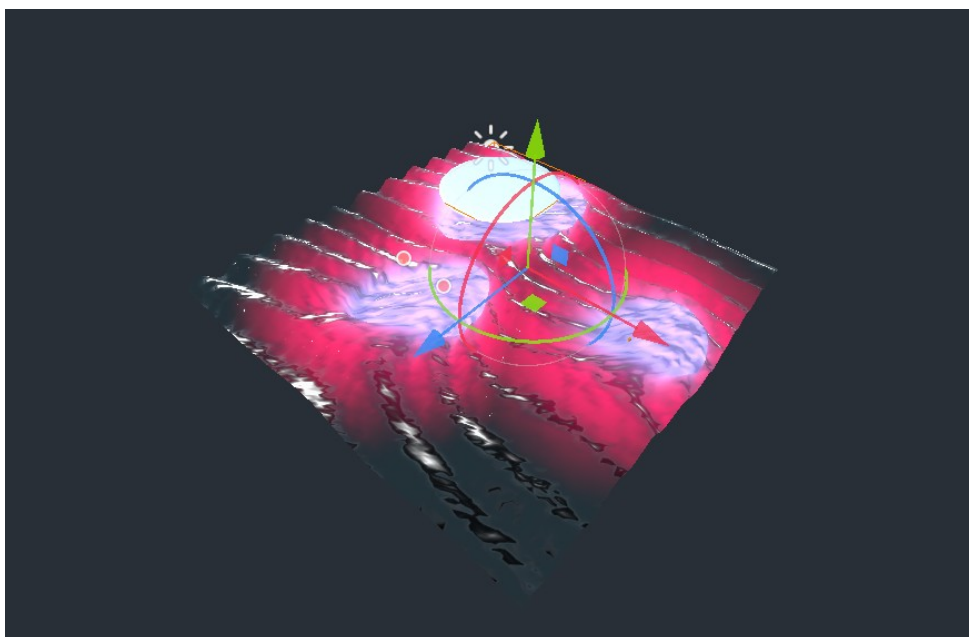
To see the influence map, navigate to **Material** → **Shader Param** → set **Show Influence Mask** to check. The red falloff signified the extend of force affecting the waves. While the inner blueish define the area of refraction, foam, and neutralized wave's height.



One thing to remember is that these point of influence nodes **must be placed under the water material hierarchy** as those depend on the parent's coordinate to do stuff. Hence, your node hierarchy should look like this.

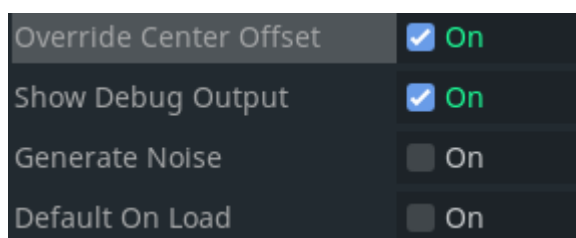


Here the example of multiple influences of three different types: **MeshInstance** with Sphere Mesh, **CollisionShape** with Cylindrical Shape, and **Position3D**.



B. Mesh Adjustment

Another functionality of this toolscript is to modify **PlaneMesh** and modify some shader parameter. We might have seen 3 other options in script variable.



Override Center Offset is checked by default and will change the offset of the **PlaneMesh** to the top left corner instead of center (+x and +z) to avoiding certain artifacts created by the influence point.

Show Debug Output is checked by default and will hide/show debug message to console.

Generate Noise is unchecked by default since all material variance has their own noise preassigned (BaseNoise.tres). But if you do check this, it will replace only the “Noise” parameter with new noise which resolution taking mesh’s size and subdivision into account.

Default On Load is unchecked by default and will reset mesh offset to the center and selected few parameters to script default.

5. Shader Parameters

Overwhelming majority of the parameters are common in all material variances, some are removed but rarely added. It’s going to be self apparent when you see it.

Name	Data Type	Description
Uv Scale	Float	Scale the noises and textures used in this material
Texture Lod	Float	Define LOD of noises and textures when read
Base Color A	Vec4	Color mix
Base Color B	Vec4	Color mix
Shallow Color	Vec4	Color of the water near the ground
Roughness	Float	Material roughness
Metallic	Float	Material metallic
Rim	Float	Fresnel amount
Hide Foam	Bool	Hide foam texture
Foam Color	Vec4	Color of foam
Foam Detail	Float	Scale foam texture
Foam Dynamic	Float	Randomize foam shapes
Foam Strength	Float	Foam color strength
Hide Refraction	Bool	Hide refraction texture
Refraction Detail	Float	Scale refraction texture
Refraction Dynamic	Float	Randomize refraction shapes
Refraction Strength	Float	Refraction color strength (always white)
Wave Speed	Float	Wave speed

Wave Length	Float	Wave length
Wave Height	Float	Wave amplitude
Wave Direction	Vec2	Direction in (+X, +Z) coordinate
Wave Turbulence	Float	Turbulence on water surface
Wave Sharpness	Float	Turbulence sharpness on water surface
Ground Visibility	Float	Visibility amount from water surface to ground
Ground Influence	Float	Scale the point influence radius
Use Ground Average	Bool	Replace default influence map distance model to averaging distance $\sum_{i=0}^n \frac{distance(\vec{ground}_i, \vec{watersurface})}{n}$ where n is total influence
Use Center Average	Bool	Replace default influence map distance model to distance to middle of all points $distance(\sum_{i=0}^n \frac{\vec{ground}_i}{n}, \vec{watersurface})$ where n is total influence
Show Influence Mask	Bool	Influence map visualization
Show Normals	Bool	show/hide normals

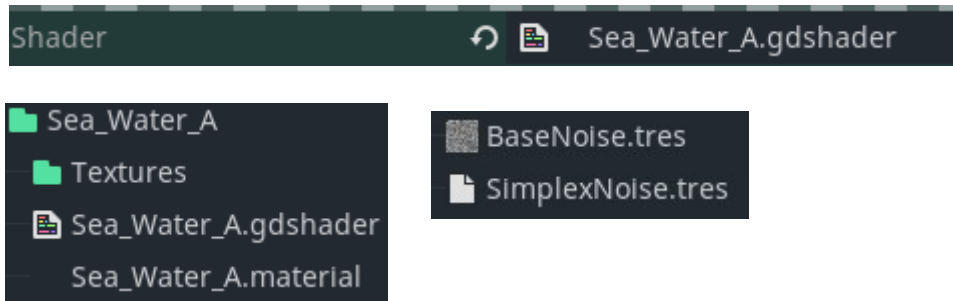
Some parameters are meant to be set by the script, it's best not to try and edit these values

Grounds Arr Length	int	
Mesh Size	Vec2	
Ground Position	Vec3	
Noise & <i>other noises</i>	Sampler2D	
Grounds Arr	Sampler2D	

6. Troubleshoot

A. When water material looks as if it has no texture or noises

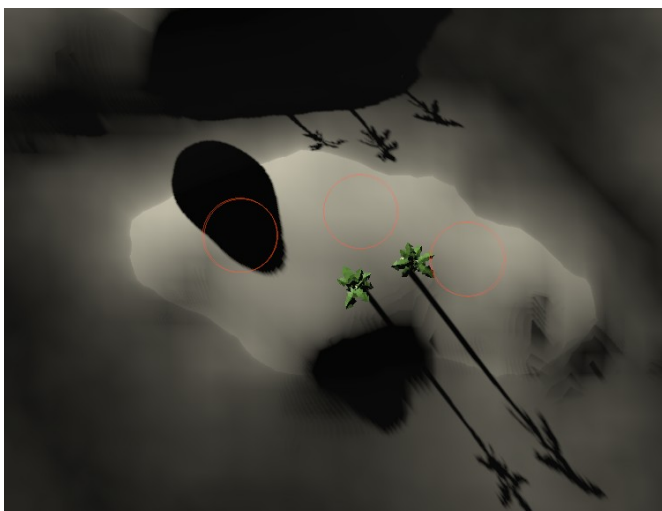
Sometimes in godot there happens where mesh or objects had lost its material dependencies. This could happen which we have no control over. However, the project is named in a such way assembling it manually can be straightforward. For example, to create a water material, you can simply drag-and-drop **gdshader** file to the **shader**. Then in **shader param** if noises are missing, assign the base normal and height texture that you can find on **Sea_Water_[X]/Texture Folder**.



B. Points of Influence wave artifacts

The power of influence map must be used sparingly and in small amount. The purpose I had in mind when making it was to hide it beneath mesh like islands with perfect circular shaped (not elliptic). So if you have a mesh that look not circular (which mostly the case), I suggest to divide it into multiple but smaller point of influences.

Finally apply **Use Ground Average** or **Use Center Average** to combine those. Of course this has limitation, as this material does not support cluster model grouping. If possible many grounds should have only one point each or one ground but many points.



C. Missing Mesh

If mesh suddenly become invisible, you can try restart the scene few times first if it's fixed. If not then surest way to prevent other problem is to remake the mesh again, then re-assign the material.